

TrapPot – AI-Powered IoT Honeypot System

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Certificate

This is for the supervisor. You only add it when you get a confirmation from your supervisor. It acts as a stamp on your document by your advisor that he/she approves it. Similar to other preamble pages this page should only be added by the end of phase-2.

Dr. Tarek Moulahi

Dedication

Write a single paragraph stating to whom you dedicate this work. This is an optional section. It should only be used by the end of phase-2 as it is abnormal to dedicate unfinished work!

Sultan Almansour, Rayan Alsamani, Faris Aljamaan, and Sulaiman Albati

Acknowledgement

On this page, you acknowledge any help you got. The help could be in any form. If your project involves financial support then you should acknowledge who supported you here. Another example is if someone helped you with providing some data or analyzing them. Any help you got should be stated here. This section is also optional. Also, the section should not be added until the end of phase-2 as it is abnormal to use it with unfinished work.

Sultan Almansour, Rayan Alsamani, Faris Aljamaan, and Sulaiman Albati

Abstract

Here you should write a clear abstract that targets four main elements. First, an introduction sentence or two that succinctly describes and appropriately connects the subject and context/background to the project's purpose. Second, two to three sentences describing the methods used to conduct the project and succinctly identifying the existing procedures, models, or programs used. Third, one or two results sentences provide a brief and specific explanation of what was discovered, accomplished, collected, or produced. Fourth, a conclusion sentence or more that provides a summarized interpretation of the results and evaluates why the results are relevant or unique compared to related projects. If the results are not optimal, a conclusion could evaluate what the project completion could lead to if more time is allowed. Finally, the abstract employs complete sentences, appropriate word choices that exclude jargon, correct grammar, conventional spelling, capitalization, and punctuation, and defines all acronyms.

Keywords: enter, keyword, here.

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Abbreviations

AI Artificial Intelligent

CNN Convolutional Neural Network

DM Data Mining

ML Machine Learning

OS Operating System

TCP Transmission Control Protocol

Chapter 1

INTRODUCTION

INTRODUCTION

DISCLAIMER: The outline given here is nothing but a suggestion. Please consult your advisor before using them.

Give an introduction to the report. What is the problem? Why is it important? Did anyone think about it before? All of this should be in brief statements. This is your opportunity to engage the reader in your report. This is also known as the hook. Recent studies outline IoT honeypot architectures and show how machine learning improves attack classification from honeypot data [2, 3].

1.1 Challenges

What are the challenges of the problem itself. Meaning, if anyone else would try to solve the same problem you are solving, what would be the challenges that they would certainly encounter? Thus the challenges are related to the domain of your project. For example, if your project is about the e-commerce, a challenge could be market penetration. You have to elaborate on each challenge you are aware of and you have to state clearly whether you're addressing the challenge or not.

1.2 Study Scope

Here you'll explain to what extent the research area will be explored, or the boundaries that the research will be performed in. Also, it should be clear what is it that you are NOT doing. For example, a project that is tackling a search algorithm could state that the search is limited to source code (a tool to search for code). Thus, you clearly communicate to the reader that you are not searching the whole web for articles, books, etc. This is just an example, you would have to add as many technical and non-technical boundaries. The clearer you are the less traps are open for the examiners.

1.3 Goals and Objectives

Here you state clearly what are your goals and what should the reader be looking for by the end of reading your report to evaluate your work. These have to be clear statements. However, when you are not sure if you can achieve a goal try to state alternatives (aka mitigations). For

example, a goal can be “by the end of our project we will reduce the search speed by 10%”. And then elaborate on that goal by for example stating how fast the best tool available now is, etc.

1.4 Report Arrangement

This section describe briefly how the reset of the report is organized. And most importantly is what should the reader expect from each one. For example, for the literature review, you could say “we go over all the established work for [project domain]. We also provide extensive background on all the essential technologies we are using in our solution”. This can be a numeric list for each chapter.

Chapter 2

LITERATURE REVIEW

LITERATURE REVIEW

DISCLAIMER: The outline given here is nothing but a suggestion. Please consult your advisor before using them.

Simple introduction to what this chapter is about.

The rest of the chapter can be organized on many different ways depending on the story you want to tell. People usually try to either go by chronological order of the publications in the field or use a classifications approach. Different approaches are also possible depending on the project domain. Below we explain two possible paths.

2.1 First Possible Outline

2.1.1 Challenges in Project Domain

Here you provide a lengthy and deep view of what are the problems defined in the past and how people before you (usually scientist) approached them whether that is using technological solution or any other type of solution. Also, you could have many references on studies and/or statistics about the issue. This section by itself can have many subsections.

2.1.2 Related Work

Here you give a description of the work that is closely related to your planned work (your planned solution). For example, if you are planning on tackling the search problem using machine learning, then in this section you should focus on papers that solve the same problem (search) using the same tools (machine learning). If the related work uses one of the tools you are planning to use but then uses more tools, then you still have to cite them here as they partially intersect with your work. In fact, you should not find a paper that does exactly what you planning on doing. If you find one then you have a problem and you have to update your goals and objectives to be novel.

2.1.3 Technical Background

Here you would have to have a subsection for any technological approach or hardware that the user might need to understand. Think of it as a reference for the reader whenever they get lost in the next chapters. For example, when in the methodology chapter you say we use divide-and-conquer algorithm, you could simply add a reference to its section here where you have

described everything about the algorithm in details. And this apply for anything that can be vague for the user and you don't want to get distracted by it when you are in upcoming chapters. Thus, you describe it here and you reference it anywhere you mention the tool/techniques.

2.2 Second Possible Outline

2.2.1 Related Work

You write in a chronological order about every effort that was made within your problem domain. You keep guiding the reader from the day when the problem was clearly defined until now and at each step you describe how each related work is relevant or irrelevant.

2.2.2 Technical Background

Here you would have to have a subsection for any technological approach or hardware that the user might need to understand. Think of it as a reference for the reader whenever they get lost in the next chapters. For example, when in the methodology chapter you say we use divide-and-conquer algorithm, you could simply add a reference to its section here where you have described everything about the algorithm in details. And this apply for anything that can be vague for the user and you don't want to get distracted by it when you are in upcoming chapters. Thus, you describe it here and you reference it anywhere you mention the tool/techniques.

Chapter 3

METHODOLOGY

METHODOLOGY

DISCLAIMER: The outline given here is nothing but a suggestion. Please consult your advisor before using them.

A research methodology outlines the systematic approach used to collect and analyze data to answer a research question or test a hypothesis. Without any heading, give a simple introduction to what this chapter is about by:

- Briefly describe the research question or hypothesis.
- Explain why the research is important.
- Provide an overview of the research methodology that will be used.

3.1 Research Design

In this section you would focus on the strategy you would follow for conducting your research. It is the blueprint for the collection, analysis, and interpretation of data to answer the research questions or test hypotheses.

- Determine the type of research design (e.g., experimental, survey, case study). In computer science we rarely use survey based studies as we can usually generate tones of data more reliably and accurately. Thus, if you plan on using a survey, please think about the justification of using it.
- Describe your research approach. This refers to the overall plan for the study, which can be quantitative, qualitative, or mixed methods.

if applicable Define the target population and sampling method.

if applicable describe your sampling method. This refers to the process of selecting participants from a population that is representative of the target population.

3.2 Data Collection

Data collection refers to the methods used to collect data, which can be primary or secondary. Primary data collection methods include surveys, interviews, observations, and experiments.

Secondary data collection methods include literature review, analysis of existing datasets, and archival research. There are essential steps that you should consider to complete this step.

- Conduct pilot tests of data collection instruments.
- Collect data from the target population.
- Ensure data quality by monitoring data collection and addressing any issues that arise.

3.3 Data Analysis

The data analysis section refers to the methods used to analyze and interpret the data collected. Quantitative data analysis methods include statistical analysis, while qualitative data analysis methods include content analysis, thematic analysis, and grounded theory analysis. Here is a list of task that you might think of.

- Organize and clean the data.
- Choose appropriate statistical or qualitative analysis methods.
- Analyze the data and report initial findings.
- Consider limitations and potential sources of bias.

3.4 Validation Plan

The validation section can describe how you plan to test your work. I.e., what you will need to validate that your solution works. And what would be the measurement of success.

3.5 Research Ethics

If applicable, the research ethics section should describe the ethical considerations in research, such as informed consent, confidentiality, privacy, and protection of vulnerable populations.

Chapter 4

IMPLEMENTATION AND TESTING

IMPLEMENTATION AND TESTING

DISCLAIMER: The outline given here is nothing but a suggestion. Please consult your advisor before using them.

Here you should write a brief introduction to this section. The introduction should not overlap with previous chapters. If there is a need to re-explain something, it should be with a purpose. Your main goal in this 1-2 page intro is to provide a high-level overview of what to expect in this chapter. Also, if you are sharing your code somewhere, you should add the link within this intro.

4.1 Programming Languages and Tools

Here you should list and explain all programming languages and tools you're using. Whether the tool/language is related to the core functionality of the project or the analysis of the project, it should be added. Avoid saying something like "we use python" it should be more of "for module x we use python because of ..." also, you should add a general statement about the tool/language and its strengths and weaknesses in general.

4.2 Software Design

Start by providing a high-level view of how the system is composed then gradually dive into details. This is a section where lots of diagrams could help. The reader of this section should have a conceptual image of how the system is developed. How many components does the system have and what are they? For each component/module explain what is the main goal of the module and link it to your project's objectives.

The use of widely known diagrams could help you get started. For example, you could use the 4+1 view models to represent your system architecture (https://en.wikipedia.org/wiki/View_model).

4.3 Algorithm(s)

Each project should have one or more essential algorithms that are crucial to the project. In this section, you should list them all (if you have more than one) with either pseudocode explaining the algorithm or using the code from one of your programming languages. A pseudocode is

preferred as the algorithm is not dependent on a given language but should reflect your code exactly. If someone implements the project using the pseudocode you provided, regardless of the language they use, it should work exactly as your project.

Please note that you don't just add pseudocode entries here. You are actually explaining it too. Thus, for each algorithm, you should have a pseudocode as a base for your explanation. We expect that you would also provide an explanation of how the algorithm fits within the whole project.

4.4 Software Testing

For a project to be reliable has to be tested. You're expected to fully or partially test your project. Also, the test could be a module-based test, an integration test, or both (preferred). Whatever you use, you should explain how you established the test cases and provide some test feedback (e.g., test coverage).

Also, the test could be related to some non-functional requirements (e.g., performance or security). If you are targeting one of these non-functional requirements then you should explain how are testing for them.

4.5 Installation

Think of this section as if you are coming back to this project 10 years from now. Hence, this section should remind you of all the steps/requirements you need to do in order to establish an instance of this project. If you are providing the project as an open-source repo, then it might be enough to explain the high-level details while providing the link here again and making sure that the repo actually has everything one needs to get started.

Chapter 5

RESULTS AND DISCUSSIONS

RESULTS AND DISCUSSIONS

DISCLAIMER: The outline given here is nothing but a suggestion. Please consult your advisor before using them.

Add an introduction here explaining what this chapter is about. The chapter is designed to cover your analysis of the problem you are solving. I.e., here you are looking at the world before you started working on this project and then after you finished working on this project, then ask yourself, how did my work affect the field I'm working in?

The ideal structure of this project is to be broken based on research questions. That is each section here is discussing your project from the point of view of some research questions. For example, if your project is about a search algorithm, then your sections could be as the following:

1. How is the accuracy of our search results compared to Google's results?
2. How fast do we return the results compared to the top 3 search engines?
3. How relevant is the result to the search topic?

Then for each one of the sections above you start analyzing your tool given the research question you posed. Also, you give a comprehensive discussion about the results. I.e., you cite each tool you are comparing to and explain how your results are better than the one provided by that other tool.

If the nature of your project doesn't fit this structure then the sections given next are the second best way to structure this chapter.

5.1 Analysis Strategy

Explain how you plan to analyze your project. What kind of questions you should ask about your work? How are you going to address these questions? E.g., are you going to run experiments to collect data? Are you going to ask users to use your system and track the usability metrics? etc?

Also, here you should highlight what are you comparing your analysis to. Are you just analyzing your work given the state of the world with and without your solution or are there established techniques that you could compare to?

5.2 Analysis Result

Start showing your results and explaining them. This section should be structurally similar to the previous sections. The analysis strategy explains what and how are we going to analyze the project and here we show the result of that analysis and explain it.

5.3 Major Findings

Here you are highlighting the major findings. The previous section should be solely for the result presentation and explaining how we can read them. Here we are giving the conclusion of that project analysis task. We mainly focus on the strengths of our project. We don't lie or fake data but, write the story of the data that favor us.

5.4 Discussion

This section is the opposite of the previous sections. In the last section, we explain the data from our point of view. Here we try to target any refutation point a reader could think of and try to address it. For example, a reader could say that your result in section x.x is not strong because of projects y and z. Your job is to think of these cases and address them ahead of the reader. Again you don't fake data, but you explain the assumptions and limitations that prevented you from comparing to tool y or being better than tool z.

Chapter 6

CONCLUSIONS AND FUTURE WORK

CONCLUSIONS AND FUTURE WORK

DISCLAIMER: The outline given here is nothing but a suggestion. Please consult your advisor before using them.

Very briefly state what this chapter is about. We say very briefly because we all know what conclusion mean. Thus, one paragraph explaining how this chapter is organized might be good enough.

6.1 Conclusion

6.1.1 Contributions

You should have stated them already in the results sections, but here you briefly re-state you project contributions. You should be as concise and clear as possible. Don't make unrealistic claims. Stay in line with what the result analysis was saying but be enthusiastic about what you made. The statements should be written in a simple paragraph form (1-3 paragraphs is good enough).

The contribution statements could be of the form "Through our solutions we were able to reduce x by a factor of y ." Then elaborate on the statement. Another could be "A user can accomplish n task in x time which is m times faster than work found in Someone et. al. [1]" and so on.

6.1.2 Limitations

Here you do the opposite of what you did in the previous sections. Any project must have some limitations. Actually, when you state the limitations it shows that you have done a thorough analysis of your work and are probably ready to solve them in future work.

However, not all limitations are solvable. A limitation could be simply a limitation forever. But stating it clearly is good.

An example of a limitation could be like "Our system can only process n requests in a second" you then elaborate on why that is a limitation and why you couldn't overcome it.

Another example could be "We only consider x type of inputs." And that is fair enough but you should state it clearly and state why you made such a choice.

6.2 Future Work

Projects are always tight on time. You cannot do everything in your mind in a short period of time. Whether that is due to priority or relevance, in this section you state what you would do if you had more time. Future work can be related to a feature you want to add or maybe a type of analysis you wish you could conduct. The more you add the more your project appears to have good potential.

Chapter 7

L^AT_EX Tutorial

L^AT_EX Tutorial

7.1 Using Lists

There are many ways to create a list. This section shows some of the most common examples and uses of lists. Remember, the sky is the limit.

7.1.0.1 Simple Bullet Point List

For simple bullet points, you use `itemize` within the begin and end tags.

- first item.
- second item.
- third item.

7.1.0.2 Bullet Point List With Different Symbols

We can change the bullet symbol as we need by defining the symbol we use for each item.

- first item.
- second item.
- third item.

7.1.0.3 Numbered List

If we want to have a numbered list we use the `\begin{enumerate} ... \end{enumerate}` keyword to have a list like the one shown below.

1. first item.
2. second item.
3. third item.

7.1.0.4 Custom Numbered List

We can define the list numbering as we want. For example, sometimes we need to have a list of specific types, like in the requirements gathering step. Hence, we say the first item should be named *REQ-1*, the second should be *REQ-2*, and so on. To get such numbered list, we define a custom label as we need as given in the example below.

REQ-1: first item.

REQ-2: second item.

REQ-3: third item.

7.1.0.5 Items Spacing

We can change the spacing between the symbol and the text and between each item as we want by changing the values of the `noitemsep`, `topsep`, `parsep`, and `partopsep`.

Note: we do not recommend changing the margin of anything as this template is designed to take care of all the formatting settings. You really need to have a good reason to change the margins.

- item 1
- item 2
- item 3

7.2 Dealing with Figures

A figure is any content wrapped between `\begin{figure}` and `\end{figure}`. A figure (or any floating object) can have different options. However, we think the most important one that we should share with you is the positioning option of the figure. Each figure can be placed somewhere on a page by adding the position option after the begin keyword `\begin{figure}[h]`. The list of positioning options and their meaning are given in Table-7.1.

Parameter	Position
h	Place the float here, i.e., approximately at the same point it occurs in the source text (however, not exactly at the spot)
t	Position at the top of the page.
b	Position at the bottom of the page.
p	Put on a special page for floats only.
!	Override internal parameters $\text{\LaTeX}$ uses for determining “good” float positions.
H	Places the float at precisely the location in the $\text{\LaTeX}$ code. Requires the float package. This is somewhat equivalent to h!.

Table 7.1: Floating objects (e.g., figure) positioning options and their meanings.

7.2.1 Simple Figure

Adding a figure simply requires you to point $\text{\LaTeX}$ to the image/pdf file location inside a `\begin{figure} ... \end{figure}` environment. The figure below (Figure-7.1) includes an image in a file located in `figures/samples/earth-in-space.jpg`. We do not resize the image or change any of its attributes. Also, we use the option `H` to place the figure right after this text.



7.2.2 Re-sized Figure

You could not see the full image above, right? That is because we did not control the image size in Figure-7.1. We will repeat the same code we used with the figure above, but this time we add the `[width=0.5\linewidth]` in front of the `\includegraphics` command. This means the new figure width (Figure-7.2) is resized to be equal to 50% of the `\linewidth` size. The `\linewidth` is defined by the document itself, and you have no control over what the `\linewidth` is. I.e., we are telling $\text{\LaTeX}$ to put the image in Figure-7.2 and make its width equal to half the allowed line width.



Figure 7.2: An example of a resized figure to be equal 50% of the defined line width. (image source [1])

7.2.3 Using Subfigures

Sometimes the figures we want to add are a collection of images instead of a single image. And we want to add all these images under a single `\begin{figure} ... \end{figure}` environment. We do this by using `\begin{subfigure} ... \end{subfigure}` inside the figure environment for each sub-figure. The code is self-explanatory, but we should point out the use of `\hfill` and `~`. We use `\hfill` to tell $\text{\LaTeX}$ to fill the rest of the horizontal space and start from the beginning of the line. Thus, we use it between Figure-7.3b and Figure-7.3c. And the symbol `~` is used to tell $\text{\LaTeX}$ that we have two distinct objects. Therefore, $\text{\LaTeX}$ will know that one has ended and another is about to start. The symbol can be seen between Figure-7.3a and Figure-7.3b. Also, between Figure-7.3c and Figure-7.3d.

Another thing we should note is the sizing configuration. With each `\begin{subfigure} ... \end{subfigure}` environment you can see that we say its size is `\{.45\textwidth\}`, which means 45% of the line width size. However, also for each `\includegraphics` command we say

`[width=\linewidth]` without any percentage value because the line width value now is already minimized to 45% of its original size as long as we are within the `\begin{subfigure} ... \end{subfigure}` environment we defined.

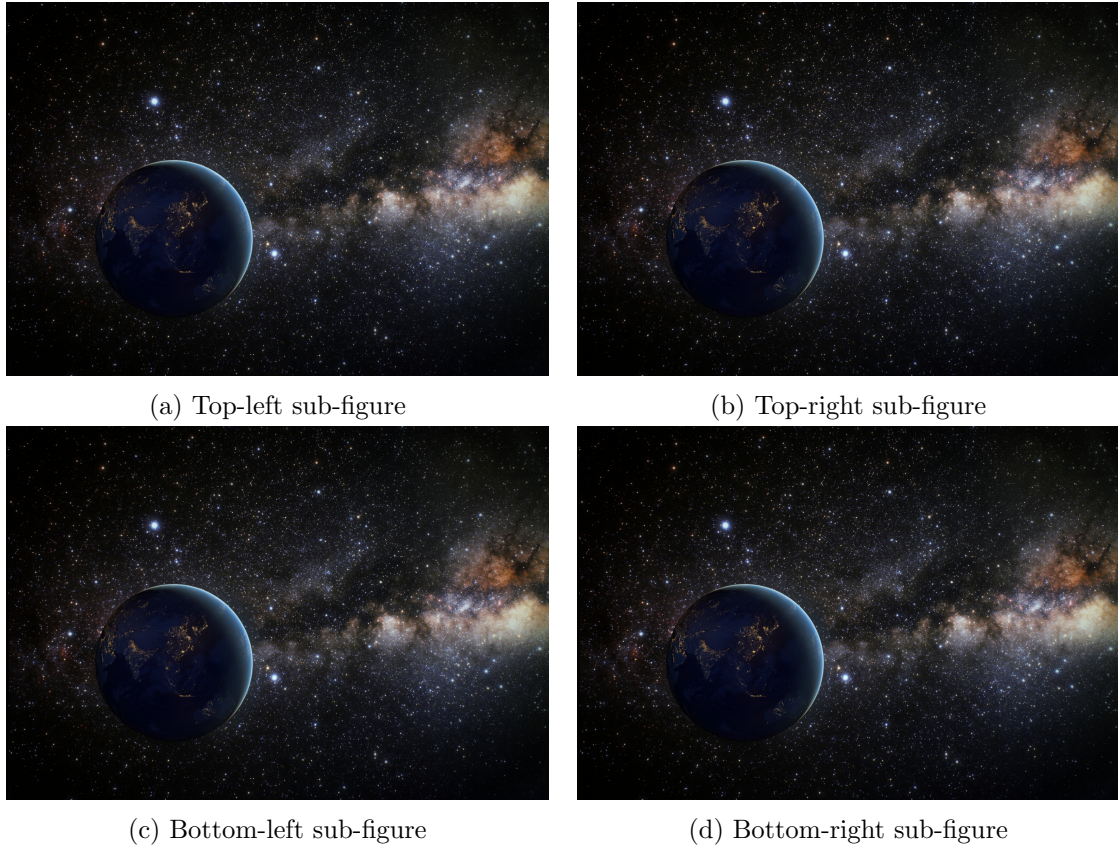


Figure 7.3: A figure that has many sub-figures placed as a grid. (image source [1])

7.2.4 Rotated Figure

Sometimes we need to make a figure show in a sideways position. For that we need to use a special floating environment named `\begin{sidewaysfigure} ... \end{sidewaysfigure}`. Figure-7.4 show an example of the use of the `sidewaysfigure` environment.



Figure 7.4: A rotated figure. (image source [1])

7.2.5 Using PNG vs. PDF Images

With the figure environment, you can either PNG/JPG or PDF-based images. You should use what you have, but whenever possible, we recommend you use a PDF-based image/chart. The reason a PDF is better than PNG (or any other fixed imaged type), is that the resolution with a PDF scales much better than other types. Take Figure-7.5 as an example and compare it to the image in Figure-7.6. If you zoom-in on the PDF-based figure you will see it scales better than the PNG-based image.

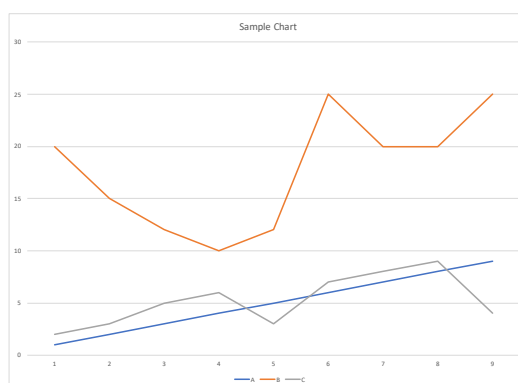


Figure 7.5: A PDF based figure.

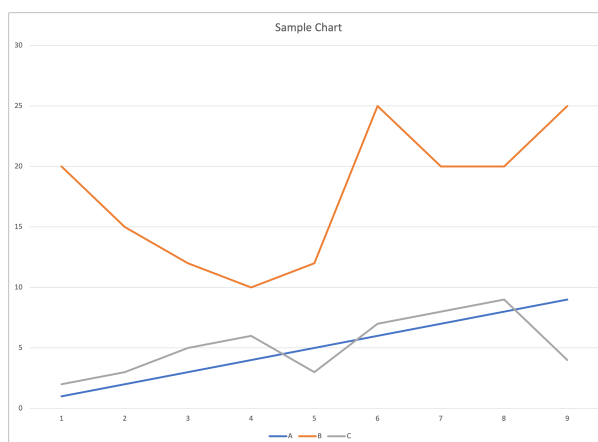


Figure 7.6: A PNG based figure.

7.3 Dealing with Tables

Tables with $\text{\LaTeX}$ are highly customizable. You can achieve any layout you want as you will see in this section. However, tables also can be intimidating the first time you use them with

L^AT_EX. Therefore, we recommend you use L^AT_EX table-building tools to start with the main body of the table and then customize it as you need. There are different tools out there, but a popular one could be <https://www.tablesgenerator.com/>.

Next, we share a few common table layouts broken by subsections as given below.

7.3.1 Simple Table

The example given in Table-7.2 is for the basic table layout. The table does not have any borders. All it has is just content broken into 3 columns and 4 rows. The table environment is given by `\begin{table} ... \end{table}`. And the content of the table itself is wrapped inside the `\begin{tabular} ... \end{tabular}` environment. Notice that after the `\begin{tabular}` we add `{l111}`, for which each letter corresponds to the column in the table, and the letter *l* in this case means align text to the left for each column.

Course Title	Grade	Level
Operating System	A	5
Data Structure	B	4
Computer Networks	A	7

Table 7.2: Simple table without borders and text align to the left.

7.3.2 All Borders Table

The table in this section (Table-7.3) is exactly the same as the table in the previous section (Table-7.2). However, this time we add all borders. For vertical lines, we updated the columns `{l111}` by adding the symbol `|` between all the letters which say we need a borderline to the left of the first column, the right of the first column, and so on. Also, we used `\hline` at the end of each row which to draw a horizontal line. With these changes only we have full borders.

Course Title	Grade	Level
Operating System	A	5
Data Structure	B	4
Computer Networks	A	7

Table 7.3: Simple table with all borders and text align to the left.

7.3.3 A Beautified Version of the Simple Table

Now you should be able to understand the basic settings that you can update to change how the table looks. In Table-7.4 we show a sample of a layout that makes the table easy to read

and looks acceptable. You can change the borders and alignment however you see fit in your case. But always think about the reader and how you could make their task of reading the tables easier.

Course Title	Grade	Level
Operating System	A	5
Data Structure	B	4
Computer Networks	A	7

Table 7.4: A beautified version of the simple table.

7.3.4 A Table with a Wrapped Text

Sometimes you would have a table with a very long text in each row. The simple tables we shared previously will go beyond the page borders as shown in Table-7.5. To make sure the table stays within the page border in the case of long text, you need to specify the table size. The trick can simply be by changing the alignments of the text on each column from general alignment as seen in Table-7.5 (`\begin{tabular}{llll}`) to be based on the page defined margins as seen in Table-7.6 (`\begin{tabular}{p{0.15\textwidth} p{0.05\textwidth} p{0.05\textwidth} p{0.6\textwidth}}`).

The new column definitions simply specify the space each column is allowed. Starting from the left, the first column is allowed 15% of the text-width, the second and third columns are allowed 5% of the text-width each, and the last column is allowed 60% of the text-width. Given these definitions, $\text{\LaTeX}$ will find a way to break the text each time it reaches the allowed size limit for each column.

Course Title	Grade	Level	Description
Operating System	A	5	Operating system course focus on the building blocks of the operating
Data Structure	B	4	The Data Structure course gives an overview of data structure concep
Computer Networks	A	7	Computer Networks course provides an introduction to computer net

Table 7.5: A table that does not handle long text.

Course Title		Grade	Level	Description
Operating System		A	5	Operating system course focus on the building blocks of the operating system, how can we secure them, and how to maintain a good performance for overall.
Data Structure		B	4	The Data Structure course gives an overview of data structure concepts, arrays, stack, queues, trees, and graphs. Discussion of various implementations of these data objects, programming styles, and run-time representations. Course also examines algorithms for sorting, searching and some graph algorithms.
Computer Networks		A	7	Computer Networks course provides an introduction to computer networks, with a special focus on Internet architecture and protocols. Topics include layered network architectures, addressing, naming, forwarding, routing, communication reliability, the client-server model, web, and email protocols.

Table 7.6: A table that wrapped long text by defining the table size.

7.3.5 A Table Spanning Multiple Pages

It is sometimes possible that you will have a table with many rows for which you cannot fit them on one page. The simple `\begin{table} ... \end{table}` environment will not help in breaking the table over multiple pages. There is a custom environment named `\begin{longtable} ... \end{longtable}` that deals with such cases. Table-7.7 shows an example of a table with 51 rows. We did not specify when the table should be broken into pages. All we did is wrapped the information inside the *longtable* environment with few definitions that make reading the table easier for the reader.

Table 7.7: A simple long-table Example

First entry	Second entry	Third entry	Fourth entry
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9
10	10	10	10
11	11	11	11
12	12	12	12
13	13	13	13
14	14	14	14
15	15	15	15
16	16	16	16
17	17	17	17
18	18	18	18
19	19	19	19
20	20	20	20
21	21	21	21
22	22	22	22
23	23	23	23
24	24	24	24
25	25	25	25
26	26	26	26
27	27	27	27
28	28	28	28

Continued on next page

Table 7.7 – *Continued from previous page*

First entry	Second entry	Third entry	Fourth entry
29	29	29	29
30	30	30	30
30	30	30	30
31	31	31	31
32	32	32	32
33	33	33	33
34	34	34	34
35	35	35	35
36	36	36	36
37	37	37	37
38	38	38	38
39	39	39	39
40	40	40	40
41	41	41	41
42	42	42	42
43	43	43	43
44	44	44	44
45	45	45	45
46	46	46	46
47	47	47	47
48	48	48	48
49	49	49	49
50	50	50	50

7.3.6 Rotated Table

In the case you have a large table that you want to contain in one page and the only solution is to have it in a side-way layout, then similar to figures, you can use `\begin{sidewaystable}` ... `\end{sidewaystable}` to put a table in a rotated layout (see Table-7.8).

Course Title	Grade	Level
Operating System	A	5
Data Structure	B	4
Computer Networks	A	7

Table 7.8: A simple example of a rotated table

7.4 Adding References

One of the essential properties of research writing or a project report is to have a well-organized document in terms of references. There are two types of reference. First, references to an outside source are called a bibliography. These are the papers, books, and webpages, you have read and would like to reference. Second, you have what is called cross-references. These are references to sections, figures, tables, etc. within your document. We explain the bibliography references in Section-7.4.1 and the cross-references in Section-7.4.2.

7.4.1 Managing & Referencing Bibliography

There are different types of publications that you can read and cite. In this tutorial, we do not dive into the details of how citations should be properly used scientifically. We only show samples of different publication types and how you can use L^AT_EX to cite them. The publication types include but are not limited to the following list:

- @article
- @book
- @collectedbook
- @conference
- @electronic
- @ieeetranbstctl
- @inbook
- @incollectedbook
- @incollection
- @injournal
- @inproceedings
- @manual
- @mastersthesis

- @misc
- @patent
- @periodical
- @phdthesis
- @preamble
- @proceedings
- @standard
- @string
- @techreport
- @unpublished

Regardless of the different types of publications, you first need to have a file where you keep all the references. In our case, the file is stored under the name `ref.bib`. If you open that file whether, in a text editor or a dedicated application for bibliography management (e.g., JabRef), you will get a list of publication information.

Each publication type has some essential information that must be added to the reference entry. However, the most essential entry for L^AT_EX to work, is the reference key. The example shown in Listing-7.1 is for a conference paper (the type is `@article`) and has a reference key set to “`Mnih:AtariRL:13`”. All the other entries are details of the reference we added. For example, the `author` key has a list of all the authors who published the paper, `title` stores the paper title, etc. These details can be different based on the reference type (e.g., article, book, inproceedings, etc).

The main takeaway from Listing-7.1 we would like to point out, is the reference key (“`Mnih:AtariRL:13`”). This is a key that you define. We recommend following a pattern to make it easier to find references. In our case, we use the first author’s last name, a hint from the paper title, and then the last two digits from the year in which the reference was published. Thus, if we put the key in a cite keyword within L^AT_EX (e.g., `\cite{Mnih:AtariRL:13}`), L^AT_EX will populate a key for it and show the reference information in the reference list [4] (click on the reference for it to take you to the reference information).


```

1 @article{Mnih:AtariRL:13,
2   author = {Volodymyr Mnih and Koray Kavukcuoglu and David Silver and Alex Graves and Ioannis Antonoglou and Daan
3     Wierstra and Martin A. Riedmiller},
4   bibsource = {dblp computer science bibliography, https://dblp.org},
5   biburl = {https://dblp.org/rec/journals/corr/MnihKSGAWR13.bib},
6   date-added = {2023-03-03 17:14:11 +0300},
7   date-modified = {2023-03-03 17:14:38 +0300},
8   eprint = {1312.5602},
9   eprinttype = {arXiv},
10  journal = {CoRR},
11  timestamp = {Mon, 13 Aug 2018 16:47:42 +0200},
12  title = {Playing Atari with Deep Reinforcement Learning},
13  url = {http://arxiv.org/abs/1312.5602},
14  volume = {abs/1312.5602},
15  year = {2013},
16  bsk-url-1 = {http://arxiv.org/abs/1312.5602}

```

Listing 7.1: Sample BibTeX Reference Entry

7.4.1.1 Citing Research Papers

The research papers can be of different types based on where are they published. The digital library you are using to obtain the paper should have the correct classification. Nevertheless, research papers are usually *@article*, *@conference*, or *@inproceedings*.

What is more important is the pieces of details that usually should not be missing. To make it easier we put these in a table (Table-7.9). Please note that you could have more details but the ones listed in the table are usually a MUST.

Attribute	Meaning
cite key	The Key you use for to cite the reference.
author	The list of authors who published the paper as they appear in the paper.
title	The paper title.
year	The year in which this paper was published.
journal or conference	The journal or conference in which the paper was published
doi	The digital identifier of an object, is the paper identifier. Note that for older papers this might not be available.
url	The link to where someone can find the paper. This could be the doi itself.

Table 7.9: Research paper essential reference properties.

We composed a sample of well-known papers to show you an example of the different citations. The first paper is for Alen Turing, one of the computer science founders. In it, Turing introduced the concept of a Turing Machine, which is a mathematical model of computers [5].

Chang et al., [6] established the foundation of the map-reduce technique that runs the internet now.

In a different subject, Volodymyr et al., [4] introduced a merge between deep learning and reinforcement learning to allow computers to teach themselves how to play games. And Sutskever et al., [7], illustrated a translation technique using neural networks.

The four papers above are a representation of different research papers from different eras. Also, they differ in being published in journals and/or conferences. We wish that you could take a look at these samples and use them as reference for any issue you might encounter.

7.4.1.2 Citing Books

Books are similar to research papers in almost everything except they do not require a link.

Table-7.10 shows the essential properties you should have in a book reference.

Attribute	Meaning
cite key	The Key you use for to cite the reference.
author	The person or persons who wrote the book.
editor	The person or persons who edited the book.
title	The name of the book.
publisher	The publisher of the book.
year	The year the book was published.
isbn	The international standard book number - a unique identifier determined by the International ISBN Agency. This is not mandatory but we highly recommend adding it.

Table 7.10: Books essential reference properties.

An example book we added is for Donald Knuth [8]. This is a book about $\text{\LaTeX}$. Listing-7.2 shows the BibTeX entry we used.

```

1 @book{Knuth:Typesetting:86,
2   author = {Donald Knuth},
3   date-added = {2023-03-04 00:49:21 +0300},
4   date-modified = {2023-03-04 00:53:48 +0300},
5   publisher = {Addison-Wesley },
6   title = {Computers and Typesetting},
7   volume = {A},
8   isbn= {978-0201734164},
9   year = {1986}
10 }
```

Listing 7.2: Sample BibTeX Reference for A Book

7.4.1.3 Citing Web-Pages

Web pages are another type of reference that you might need to add. An essential different in citing web pages is to indicate when did you visit the web pages. Because web pages are subject to change, you need to record when did you visit the page and obtained the information you rely on.

Another difference when dealing with webpages is that they do not have an explicit type in the BibTex referencing system. Most commonly, people use the type *@misc* to add a reference to a webpage. Table-7.11 shows the essential properties you have to add when citing a webpage.

Attribute	Meaning
cite key	The Key you use for to cite the reference.
author	The person or persons who published the web page. This sometimes might be difficult to find if the web page is owned by an organization.
title	The title of the web page.
howpublished	How was this misc reference publish? For websites we add the webpage url.
note	To add the date you accessed the webpage.

Table 7.11: Webpage essential reference properties.

As an example, we used our college webpage [9] as a reference. The entry used for the citation used is given in Listing-7.3.

```

1 @misc{coc:23,
2   author = {Qassim University},
3   date-modified = {2023-03-04 01:12:47 +0300},
4   note = {Accessed: 2023-03-04},
5   title = {College of Computer at Qassim University},
6   url = {https://coc.qu.edu.sa/},
7   bdsk-url-1 = {https://coc.qu.edu.sa/}
8 }
```

Listing 7.3: Sample BibTex Reference for A Webpage

7.4.2 Cross Referencing

Cross-referencing is much easier than referencing a bibliography. You only need two things.

1. Add a label to any part in the document you want to reference. Labels are easily added by using the keyword `\label{your_key}`.
2. Reference the label wherever you need to by using the keyword `\ref{your_key}`.

Here are some cross-references that you can test.

- This is a reference to Section-7.1 where we talked about how to use lists.
- This is a reference to Figure-7.3 where we showed how to use sub-figures.
- This is a reference to Table-7.4 where we showed a simple table example.

7.5 Mathematical Mode

One of the well-known strengths of $\text{\LaTeX}$ is the ease and comprehensibility of writing mathematical notations. In this section, we show the two cases of writing in mathematical mode. (Inline as shown in Section-7.5.1 and equations as shown in Section-7.5.2).

Please note that we do not cover all the possible symbols and math forms in this section. This is just a tutorial to get you started. A good reference for mathematical writing can be found in <https://www1.cmc.edu/pages/faculty/aaksoy/latex/latexthree.html>.

7.5.1 Using Mathematical Notations

Writing mathematical notation inline with $\text{\LaTeX}$ requires you to wrap the mathematical symbols between one of the following wrappers.

- `\( ... \)`
- `$ ... $`
- `\begin{math}...\end{math}`

```
1 These are examples of inline math equations. This first example, $\sqrt{3}$, uses dollar signs to write an equation
  and it shows a square root. The second example here, \(E=mc^2\), uses parentheses to write an equation and shows
  the mass-energy equivalence formula. The last example we show, \begin{math} P: \widehat{G} \rightarrow G \end{math}, uses {\LaTeX} math keyword to write a G with a wide hat and an arrow symbol.
2
3 Regardless of the wrapper, you use the result will always be the same.
```

Listing 7.4: Sample Paragraph with Inline Math Equations

The $\text{\LaTeX}$ input shown in Listing-7.4 above will result in the text shown below when we typeset it.

These are examples of inline math equations. This first example, $\sqrt{3}$, uses dollar signs to write an equation and it shows a square root. The second example here, $E = mc^2$, uses parentheses to write an equation and shows the mass-energy equivalence formula. The last

example we show, $P : \widehat{G} \rightarrow G$, uses L^AT_EX math keyword to write a G with a wide hat and an arrow symbol.

Regardless of the wrapper, you use the result will always be the same.

7.5.2 Writing Mathematical Equations

The other mathematical mode is available when you want to write a stand-alone mathematical formula/equation. For stand-alone math mode, we use the wrapper `\begin{equation} ... \end{equation}`.

$$\sum_{n=1}^{\infty} 2^{-n} = 1 \quad (7.1)$$

$$\int_a^b x^2 dx \quad (7.2)$$

$$f(x) = \frac{P(x)}{Q(x)} \quad \text{and} \quad f(x) = \frac{P(x)}{Q(x)} \quad \text{and} \quad f(x) = \frac{P(x)}{Q(x)} \quad (7.3)$$

Here are three mathematical equations samples. Note that we can reference them just like figures and tables. I.e., the equations are Equation-7.1, Equation-7.2, and Equation-7.3.

7.6 Acronyms

L^AT_EX provides a nice package that helps you manage acronyms. The package is defined in the `00-main.tex` file. All the acronyms are defined in the main file (`00-main.tex`) to be accessible for all other chapters. As an example, we declared the acronyms shown in Listing-7.5.

```

1 \DeclareAcronym{os}{
2   short = OS ,
3   long  = Operating System ,
4   tag   = abbrev
5 }
6 \DeclareAcronym{ai}{
7   short = AI ,
8   long  = Artificial Intelligent ,
9   tag   = abbrev
10 }
11 \DeclareAcronym{dm}{
12   short = DM ,
13   long  = Data Mining ,
14   tag   = abbrev
15 }
16 \DeclareAcronym{tcp}{
17   short = TCP ,

```

```
18   long  = Transmission Control Protocol ,
19   tag   = abbrev
20 }
21 \DeclareAcronym{ml}{
22   short = ML ,
23   long  = Machine Learning ,
24   tag   = abbrev
25 }
26 \DeclareAcronym{cnn}{
27   short = CNN ,
28   long  = Convolutional Neural Network ,
29   tag   = abbrev
30 }
```

Listing 7.5: Sample Acronyms Declarations

Now that the acronyms are declared, we can easily refer to them whenever we write about them in text. Here are the acronyms we declared in use Operating Systems (OS), Artificial Intelligent (AI), Transmission Control Protocol (TCP), Machine Learning (ML), Convolutional Neural Network (CNN), and Data Mining (DM).

Appendix A

Project Installation

References

- [1] P. D. Pictures, “Earth in space,” <https://www.publicdomainpictures.net/en/view-image.php?image=308781&picture=earth-in-space>, accessed: 2023-03-02.
- [2] M. Alawadi, M. Alshammari, and S. Al-Hejaili, “A comprehensive study on iot honeypots: Models, taxonomies, tools, and future directions,” *IEEE Internet of Things Journal*, vol. 9, no. 18, pp. 16 983–17 000, September 2022.
- [3] V. S. S. Y. K. R. M. S. G. U. S., “Machine learning-based attack classification using honeypot data,” *International Journal of Information Security*, vol. 20, no. 4, pp. 501–518, 2021.
- [4] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, and M. A. Riedmiller, “Playing atari with deep reinforcement learning,” *CoRR*, vol. abs/1312.5602, 2013. [Online]. Available: <http://arxiv.org/abs/1312.5602>
- [5] A. M. Turing, “On computable numbers, with an application to the entscheidungsproblem,” *J. of Math*, vol. 58, no. 345-363, p. 5, 1936. [Online]. Available: <https://www.wolframscience.com/prizes/tm23/images/Turing.pdf>
- [6] F. Chang, J. Dean, S. Ghemawat, W. C. Hsieh, D. A. Wallach, M. Burrows, T. Chandra, A. Fikes, and R. E. Gruber, “Bigtable: A distributed storage system for structured data,” *ACM Trans. Comput. Syst.*, vol. 26, no. 2, jun 2008. [Online]. Available: <https://doi.org/10.1145/1365815.1365816>
- [7] I. Sutskever, O. Vinyals, and Q. V. Le, “Sequence to sequence learning with neural networks,” in *Proceedings of the 27th International Conference on Neural Information Processing Systems - Volume 2*, ser. NIPS’14. Cambridge, MA, USA: MIT Press, 2014, pp. 3104–3112. [Online]. Available: <http://arxiv.org/abs/1409.3215>
- [8] D. Knuth, *Computers and Typesetting*. Addison-Wesley, 1986, vol. A.

- [9] Q. University, “College of computer at qassim university,” <https://coc.qu.edu.sa/>, accessed: 2023-03-04.